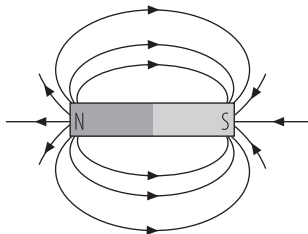


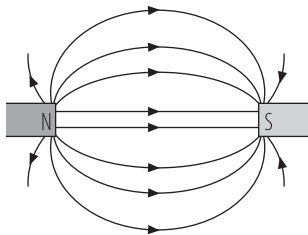
Unit 1: Answers

1.
 - a) Magnetic field
 - b) (Permanent) magnet
 - c) Geographical north
 - d) Compass
 - e) Aurora
2.
 - a) Like poles (N and N or S and S) repel; opposite poles (N and S) attract.
 - b)
 - i) Repulsive
 - ii) Repulsive
 - iii) Attractive

c)



d)



- e) Field (non-contact) force; magnets do not have to be in contact to exert forces on each other.

f)



Unit 2: Answers

1.
 - a) Neutral / uncharged
 - b) $-1,6 \times 10^{-19} \text{ C}$
 - c) Repulsive
 - d) Positive
 - e) Electric field
 - f) Principle of Charge Quantisation
 - g) Principle of Conservation of Charge

2.
 - a) Negative
 - b) Because the balloon and rod repel each other, they must have like charges. If the rod is negatively charged, the balloon must also be negative.
 - c) Charging by friction / triboelectric charging
 - d) Positive
 - e) Charge transferred:
 $Q = nq_e$
 $= 300 \times 1,6 \times 10^{-19} \text{ C} = 4,8 \times 10^{-17} \text{ C}$
 Balloon gains electrons: $-4,8 \times 10^{-17} \text{ C}$
 Hair loses electrons: $+4,8 \times 10^{-17} \text{ C}$

3.
 - a) Principle of Charge Quantisation
 - b) i) Number of excess electrons:
 $n = \frac{Q}{q_e}$
 $= \frac{-8 \times 10^{-17} \text{ C}}{-1,6 \times 10^{-19}} = 500 \text{ electrons}$
 - ii) Number of electrons present if object was neutral:
 $n = \frac{Q}{q_e}$
 $= \frac{-4,8 \times 10^{-17} \text{ C}}{-1,6 \times 10^{-19}} = 300 \text{ electrons}$
 Total number of electrons = $300 + 500$
 $= 800$
 - c) $Q = nq_p$
 $= 235 \times +1,6 \times 10^{-19} \text{ C}$
 $= +3,76 \times 10^{-17} \text{ C}$
 - d) i) $Q = nq_p$
 $= 600 \times +1,6 \times 10^{-19} \text{ C}$
 $= +9,6 \times 10^{-17} \text{ C}$
 - ii) $\frac{Q_A + Q_B}{2} = \frac{-5,6 \times 10^{-17} \text{ C} + 9,6 \times 10^{-17} \text{ C}}{2}$
 $= +2 \times 10^{-17} \text{ C}$

- iii) Sphere A changed its charge from $-5,6 \times 10^{-17} \text{ C}$ to $+2 \times 10^{-17} \text{ C}$, which is a change of $-7,6 \times 10^{-17} \text{ C}$. Sphere B changed its charge from $+9,6 \times 10^{-17} \text{ C}$ to $+2 \times 10^{-17} \text{ C}$, which is a change of $-7,6 \times 10^{-17} \text{ C}$. Number of electrons transferred:

$$n = \frac{Q}{q_e}$$

$$= \frac{-7,6 \times 10^{-17}}{-1,6 \times 10^{-19} \text{ C}} = 475 \text{ electrons}$$

- e) i) P to Q
- ii) $\frac{Q_P + Q_Q}{2} = \frac{2,4 \times 10^{-17} \text{ C} + 3,2 \times 10^{-17} \text{ C}}{2}$
 $= +2,8 \times 10^{-17} \text{ C}$
 P changed its charge from $+2,4 \times 10^{-17} \text{ C}$ to $+2,8 \times 10^{-17} \text{ C}$, which is a change of $+4 \times 10^{-18} \text{ C}$. Q changed its charge from $+3,2 \times 10^{-17} \text{ C}$ to $+2,8 \times 10^{-17} \text{ C}$, which is a change of $-4 \times 10^{-18} \text{ C}$.

- iii) Number of electrons transferred:

$$n = \frac{Q}{q_e}$$

$$= \frac{-4 \times 10^{-18} \text{ C}}{-1,6 \times 10^{-19} \text{ C}} = 25 \text{ electrons}$$

Unit 3: Answers

1.
 - a) Electrons
 - b) Joule
 - c) Emf
 - d) Parallel
 - e) Series
 - f) Electric current
 - g) Potential difference / voltage
 - h) 1 ohm

2.
 - a) The bulb in circuit B will burn brighter.
Circuit B has two cells in series that will provide twice the amount of emf and therefore twice the current through the bulb.
 - b) i) The direction of current is from the positive end of the battery and through the circuit to the negative end of the battery.
ii) The rate of flow of charge
 - c) i) 0,5 C of charge pass a point in the circuit in 1 s.
ii) The total resistance will increase and that will result in a decrease in current.
 - d) i) A battery goes flat when all its chemical potential energy has been converted into other forms of energy (heat and light).
ii) Rechargeable battery
iii) Cell phones, laptops, cameras

3.
 - a)
$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$$

$$= \frac{1}{6\ \Omega} + \frac{1}{12\ \Omega} + \frac{1}{36\ \Omega} + \frac{1}{18\ \Omega}$$

$$= \frac{6 + 3 + 1 + 2}{36}\ \Omega = \frac{12}{36}\ \Omega$$

$$R_p = 3\ \Omega$$
 - b) i) 1,5 A
12 Ω is twice the resistance of 6 Ω .
Therefore the current will be half of the current through the 6 Ω resistor.
ii) 0,5 A
36 Ω is 6 times the resistance of 6 Ω .
Therefore the current will be $\frac{1}{6}$ of the current through the 6 Ω resistor.

- iii) 1 A
18 Ω is 3 times the resistance of 6 Ω .
Therefore the current will be $\frac{1}{3}$ of the current through the 6 Ω resistor.
 - c) $3 + 1,5 + 0,5 + 1 = 6\ \text{A}$
 - d) i) R_T will increase.
ii) I_T will decrease.
 - e) $Q = I\Delta t = 6\ \text{A} \times 2 \times 60\ \text{s} = 720\ \text{C}$

4.
 - a) 3V
 - b) 2,8 V
 - c) $W = VQ$
 $= 2,8\ \text{V} \times 7 \times 10^{-9}\ \text{C} = 1,96 \times 10^{-8}\ \text{J}$
 - d) i) Total current / current through the battery / current through $R_1 + R_2$
ii) Current through R_2
iii) Current through R_1
 - e) $V_{R_1} = V_{R_2}$